

Will Fisher and Michael Munoz

Tasfia Mashiat

Topics in Computer Science I:

Ethics in Artificial Intelligence

01 May 2026

Fairness and Performance Analysis of Machine Learning Models Across Multiple Datasets

Abstract

Machine learning models have become integral to many decision-making systems; however, they tend to generate results that could adversely affect specific populations. In this experiment, three classification models, Logistic Regression, Random Forest, and XGBoost, are analyzed based on their performance and fairness in two different datasets, one being a synthetic dataset for healthcare patients and the other a diversity dataset in the tech industry. The difference in structure and demographics between the datasets allows the researchers to study the effects of differences in datasets on model performances. Both performance indicators (accuracy, precision, recall, F1-score, ROC-AUC) and fairness indicators (demographic parity, equal opportunity, equalized odds, disparate impact) were used to determine the performance and fairness of each model on both datasets. The results indicate that some models exhibit high levels of accuracy but generate significant levels of bias based on the dataset used.

Introduction

Modern data-driven decision-making systems rely heavily on machine learning algorithms for predictions. While machine learning algorithms are known for high prediction accuracy, they suffer from one important flaw – biases. Due to the training process involving real-world datasets, machine learning models tend to reproduce inequalities existing within such data. Thus, there is a growing concern about fairness of machine learning predictions and classifications.

Machine learning bias appears in cases when models produce systematically different results depending on some demographic criteria. For example, in health care, this leads to discriminatory treatment, and in workforce analytics, such biases may exacerbate inequality between demographic groups by reinforcing certain practices in hiring and promotion. In order

to be applied ethically, machine learning systems require careful analysis for detecting potential biases.

In this study, the effects of dataset features on performance and fairness of machine learning models are assessed. To illustrate how bias emerges due to variations of datasets' properties, this paper uses two separate datasets and evaluates performance of three popular classifiers, applying the same measures to estimate performance and fairness.

Methods

Dataset 1: Synthetic Healthcare Patient Journey Dataset

Source: <https://www.kaggle.com/datasets/emirhanakku/synthetic-healthcare-patient-journey-dataset>

The dataset of the Synthetic Healthcare Patient Journey represents a structured dataset that can be used to simulate the experiences of patients undergoing medical procedures. The records include all relevant information on the journey of patients from admittance to discharge, and in some cases, re-admittance.

The synthetic dataset contains various features describing different characteristics, including demographic features and hospital-related characteristics as well as outcomes. The features include variables describing the age and gender of patients, admission type, department, time spent in hospital, and treatment intensity. The outcomes represent variables such as readmission status, patient satisfaction, and possible complications.

The dataset can be used for multiple predictive modeling purposes and specifically for binary classification problems. It can be used for predicting outcomes based on various patient and hospital factors. Despite the fact that it is synthetic, it provides relevant data that allows one to analyze the impact of various factors without the risks associated with sensitive personal data.

Dataset 2: Diversity in Tech Companies Dataset

Source: <https://www.kaggle.com/datasets/jainaru/diversity-in-tech-companies/data>

The Diversity in Tech Companies Dataset includes information on the demographics of employees at various tech companies. In particular, it addresses the topic of diversity in connection with gender and race.

The information in this dataset is mostly organized on a company-by-company basis, with each observation corresponding to demographic distributions within a company for a certain year. Notable attributes include proportions or counts of employees according to gender and race (White, Asian, Black, Hispanic).

One thing that is notable about this dataset is that it is not made up of observations on an individual level, like the dataset about health care. This difference affects its use in ML fairness considerations. The dataset is especially valuable for analyzing the disparities in employee representation in the technology sector.

Approach of Combined Analysis. The models were individually tested and trained on the same datasets with identical parameter sets. This ensures that the results obtained can be directly compared and analyzed for differences arising from variations in dataset characteristics.

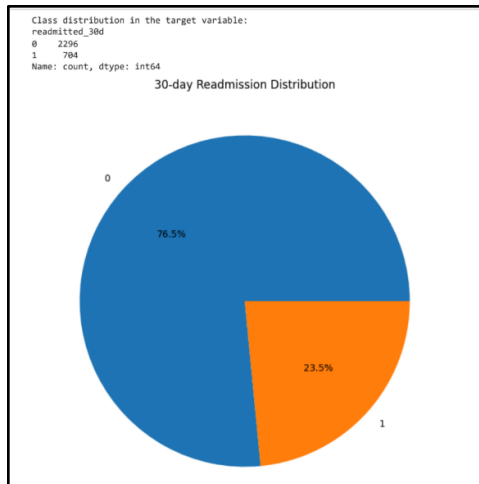
Data Preprocessing. Preprocessing of the data took place in order to make it suitable for modelling. Missing values were properly addressed and categorical variables were numerically encoded and normalized if required. Each dataset was split into training and testing sets following the same proportions.

Models Selected. There are three models utilized in this research: Logistic Regression, Random Forest Classifier, and XGBoost. The last two are more complex algorithms compared to Logistic Regression. All the models have been tested with the same parameters.

Performance Metrics. Metrics chosen to evaluate the model include accuracy, precision, recall, F1-score, and AUC value. These measures allow for determining general model effectiveness and the type of its errors.

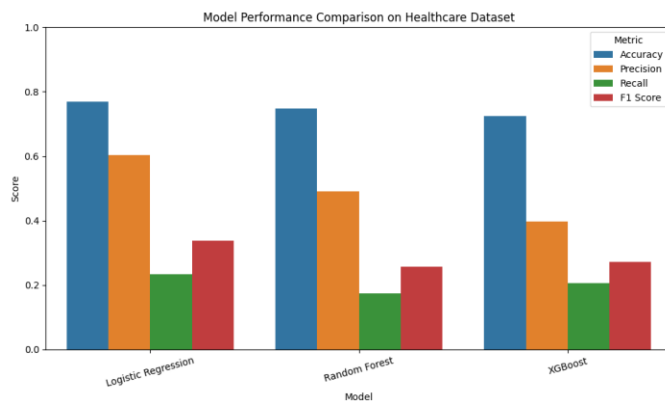
Results on Healthcare Dataset

The healthcare dataset showed relatively strong and consistent performance across all three models. Before evaluating the models, it is important to understand the distribution of the target variable. As shown below, the dataset is somewhat imbalanced, with a larger proportion of patients not being readmitted within 30 days.

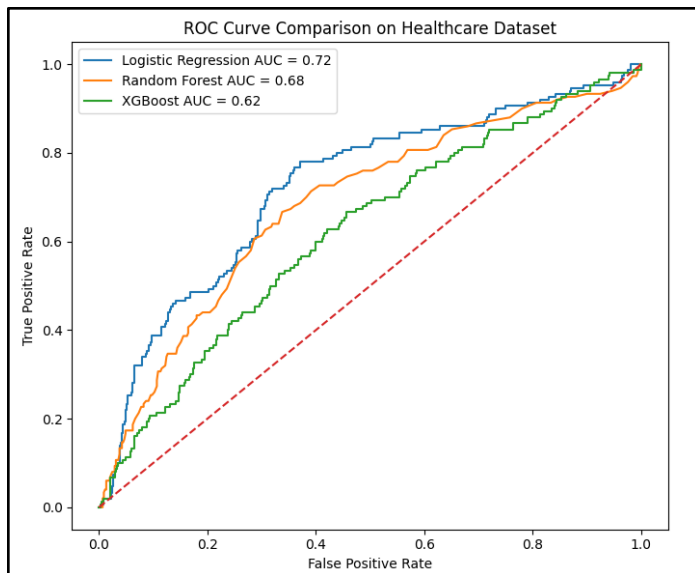


Approximately 76.5% of patients were not readmitted, while 23.5% were readmitted. This imbalance helps explain why models achieved higher accuracy but lower recall, as predicting the majority class is easier.

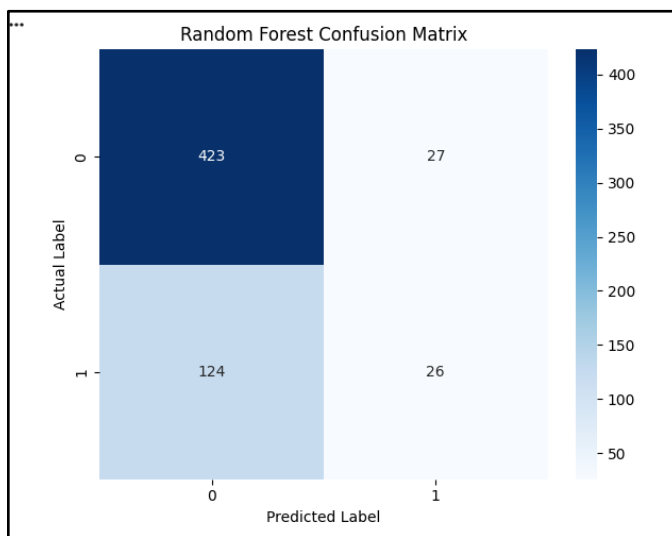
When comparing model performance, all three models achieved relatively high accuracy. Logistic Regression slightly outperformed the ensemble models in this case, although the differences were not large. Precision was highest for Logistic Regression, while recall remained low across all models, indicating difficulty in correctly identifying readmitted patients.



The ROC curve further supports these findings. Logistic Regression achieved the highest AUC (0.72), followed by Random Forest (0.68) and XGBoost (0.62). This indicates that Logistic Regression performed best overall at distinguishing between classes on this dataset.

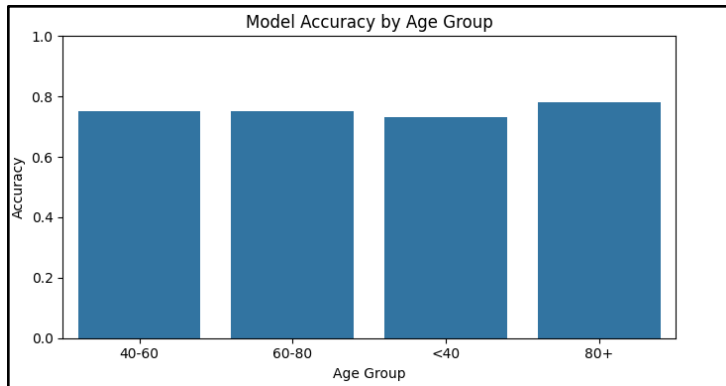


To better understand prediction behavior, the confusion matrix for the Random Forest model is shown below.



The confusion matrix shows that the model correctly predicted a large number of non-readmitted cases but struggled with identifying readmitted patients. There were significantly more false negatives (124) than true positives (26), reinforcing the observation that recall is low.

Fairness was evaluated by examining performance across demographic groups. The chart below shows model accuracy by age group.

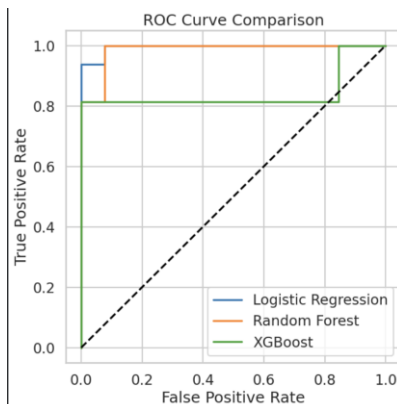


Accuracy was relatively consistent across most age groups, although slightly higher for the 80+ group. While differences are not extreme, this still suggests small variations in model performance across groups.

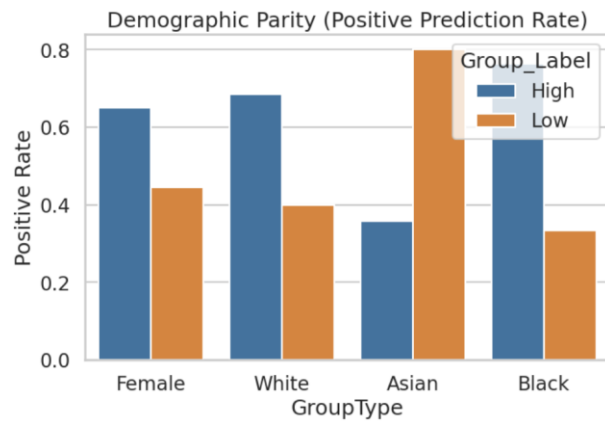
The healthcare dataset produced stable model performance with moderate fairness differences. However, the class imbalance and low recall highlight limitations in identifying high-risk patients.

Results on Diversity Dataset

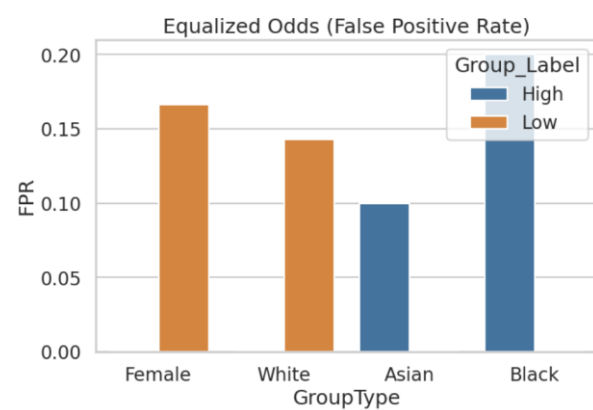
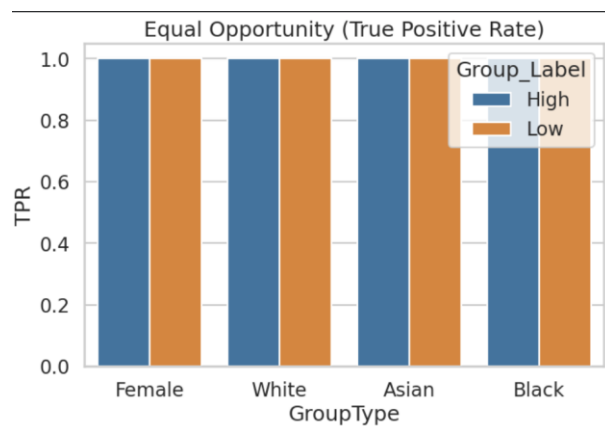
The diversity dataset produced strong classification performance across all three models, with Logistic Regression achieving the highest ROC-AUC, followed by Random Forest and XGBoost. This shows that the models were effective at predicting whether a company had high Latino representation. However, similar to the healthcare dataset, strong performance did not guarantee fairness.



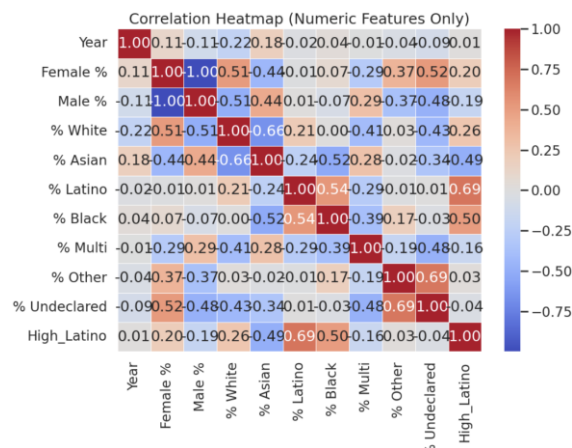
Fairness analysis revealed significant disparities across demographic groups, particularly for Asian and Black representation. The model violated demographic parity, as there were clear differences in positive prediction rates between high and low groups. Disparate impact values were also below the commonly accepted threshold of 0.8, confirming the presence of bias.



Although the model satisfied equal opportunity with consistent true positive rates across groups, differences in false positive rates indicated violations of equalized odds.



To further understand model behavior, confusion matrices were examined across groups. The results show that some groups had more incorrect predictions than others, reinforcing the presence of bias in model outcomes.



To address these disparities, several bias mitigation techniques were applied, including suppression, reweighing, sampling, and massaging. Suppression was largely ineffective, showing that removing sensitive attributes alone does not eliminate bias due to proxy variables. Reweighting provided limited improvement. However, sampling significantly reduced bias for Black representation while also improving accuracy, and massaging reduced bias for Asian representation, though with a slight decrease in performance.

	Protected Group	Method	Accuracy	Positive Rate Gap	Disparate Impact
4	Asian	Massaging	0.620690	0.176190	0.669643
2	Asian	Reweighting	0.655172	0.242857	0.595238
3	Asian	Sampling	0.655172	0.242857	0.595238
0	Asian	Baseline	0.758621	0.442857	0.446429
1	Asian	Suppression	0.758621	0.442857	0.446429
8	Black	Sampling	0.896552	0.313725	0.515152
6	Black	Suppression	0.931034	0.490196	0.404762
9	Black	Massaging	0.724138	0.490196	0.404762
7	Black	Reweighting	0.793103	0.799020	0.094444
5	Black	Baseline	0.758621	0.857843	0.088542

These results show that bias in machine learning models is strongly influenced by the dataset structure and feature relationships, and that preprocessing techniques can help reduce disparities, though trade-offs may occur.

Discussion

Results Examination. Variation in results across datasets indicates that the structure and distribution of data has a profound influence on the outcomes of models. A model that works well on one dataset might not work well on another.

Remarks on Fairness. Fairness analysis showed differences between demographic groups, especially in the diversity dataset. The differences are likely due to the imbalanced representation and the structure of the dataset. The results from the diversity dataset further reinforce that fairness issues are not solely dependent on the model but are deeply rooted in the dataset itself.

Ethical Statement. Bias in machine learning models have adverse consequences in the real world. In health care, bias can influence decisions on treatments and in workforce analysis, it can perpetuate inequality. Fairness is needed for ethical deployment. The results of the diversity dataset study show that models can produce unequal outcomes across demographic groups even when overall performance is strong. This highlights the importance of carefully evaluating and mitigating bias to avoid reinforcing existing disparities.

Weaknesses. The limitation of this study is synthetic and aggregated datasets. These datasets may not reflect the complexity of the real world. Also, only a limited set of models and metrics were considered. Another limitation is that some of the data is grouped rather than based on individual records, which limits how detailed the analysis can be. Additionally, some groups have fewer data points, which can make differences seem larger and affect the reliability of the fairness results.

Strategies to Decrease Bias. To reduce bias in machine learning models. These practices should be followed. First, careful data collection and preprocessing are big, as bias often originates from imbalanced or unrepresentative datasets. Techniques such as reweighing, sampling, and data massaging can help mitigate disparities before training. Next, models should be evaluated using multiple fairness metrics, since no single metric fully captures fairness. Then, sensitive attributes should be handled cautiously, as simply removing them does not eliminate bias due to proxy variables. Lastly, continuous monitoring and evaluation are important to make sure that models remain fair when used.

Conclusion

This study shows that the performance and fairness of machine learning models are strongly dependent on dataset characteristics. Some models can achieve high accuracy, but still produce biased results depending on the dataset. Comparison of results from two datasets demonstrates the need to test models in different contexts. Fairness in machine learning requires careful considerations in data and model design.

References

- <https://www.kaggle.com/datasets/emirhanakku/synthetic-healthcare-patient-journey-dataset>
- <https://www.kaggle.com/datasets/jainaru/diversity-in-tech-companies/data>